

A SOTA-inspired 3D Scene Completion Prototype Using Real NYUv2 Depth Data

Abstract

This project implements a complete 3D scene completion prototype using real NYUv2 depth data. The system converts NYUv2 depth maps into point clouds, voxelizes them into 3D occupancy volumes, constructs incomplete input occupancy and proxy target occupancy, trains a residual-attention 3D U-Net, and evaluates the predicted occupancy using IoU, Precision, Recall, and F1 score. The work is positioned as a SOTA-inspired prototype rather than a public-leaderboard SOTA method.

1. Introduction

3D scene completion aims to infer missing or unobserved occupancy from partial observations. Modern scene completion methods are often based on volumetric neural networks, encoder-decoder architectures, residual connections, attention modules, and losses designed for sparse occupancy. This coursework project focuses on building an end-to-end, reproducible prototype using real depth maps from NYUv2.

The key goal is not to reproduce a large-scale benchmark result, but to demonstrate a credible pipeline from real RGB-D data to 3D occupancy prediction.

2. Dataset

The project uses the official NYU Depth V2 labeled dataset. NYUv2 contains indoor scenes captured using Microsoft Kinect, including aligned RGB and depth images. The labeled subset contains 1449 densely labeled RGB-D pairs. The data file also includes raw depth maps and preprocessed depth maps with missing values filled.

In this project:

- `rawDepths` is used to construct incomplete input occupancy.
- `depths` is used to construct the target/proxy occupancy.

This keeps both input and target/proxy occupancy derived from real NYUv2 depth data.

3. Problem formulation

Given an incomplete input occupancy volume X , the model predicts a completed occupancy probability volume $\hat{Y}$. The proxy target Y is derived from NYUv2 in-painted depth maps. The task is binary occupancy prediction over a fixed 3D voxel grid.

It is important to note that NYUv2 single-view depth does not directly provide complete physical 3D ground truth. Therefore, the project evaluates against a depth-derived proxy target rather than claiming full scene-completion ground truth.

4. Method

4.1 Depth to point cloud

For every valid depth pixel, the point cloud is computed with the pinhole camera model:

$$\begin{aligned} X &= (u - c_x) * Z / f_x \\ Y &= (v - c_y) * Z / f_y \\ Z &= \text{depth}(u, v) \end{aligned}$$

Invalid, zero, NaN, and out-of-range depth values are removed.

4.2 Voxelization

The point cloud is clipped to fixed camera-space bounds and voxelized into a binary occupancy grid. The default grid resolution is $64 \times 64 \times 64$, stored in $[D, H, W] = [z, y, x]$ order.

4.3 Input and proxy target construction

The input occupancy is constructed from raw NYUv2 depth maps. The target/proxy occupancy is constructed from in-painted NYUv2 depth maps. During training, optional voxel dropout and cuboid masks remove occupied voxels from the input volume only, simulating additional incompleteness while keeping the target derived from real NYUv2 depth.

4.4 Network architecture

The model is a lightweight residual-attention 3D U-Net. It contains:

- residual 3D convolutional blocks,
- downsampling encoder path,
- squeeze-and-excitation attention modules,
- upsampling decoder path,
- skip connections,
- one-channel occupancy logits output.

This architecture is SOTA-inspired because it borrows common design principles from strong volumetric completion models, but it is intentionally reduced for coursework-scale training.

4.5 Loss function

The training objective combines binary cross-entropy with logits and Dice loss:

$$\text{Loss} = \text{BCEWithLogitsLoss} + \text{lambda} * \text{DiceLoss}$$

A positive class weight is used because occupied voxels are sparse relative to empty voxels.

5. Evaluation

Predicted probabilities are thresholded to binary occupancy. The following metrics are computed:

- IoU
- Precision
- Recall
- F1 score

These metrics are computed over the proxy target occupancy volume.

6. Visualization

The project saves side-by-side 3D visualizations for:

1. input occupancy,
2. predicted occupancy,
3. target/proxy occupancy.

This helps verify whether the model reconstructs missing observed surfaces and produces outputs structurally closer to the proxy target.

7. Expected experiments

Suggested experiments include:

1. raw depth input versus in-painted depth proxy target,
2. attention enabled versus disabled,
3. BCE only versus BCE + Dice loss,
4. different input masking strengths,
5. grid resolution comparison between 32^3 and 64^3 .

8. Limitations

This project has several limitations:

- NYUv2 does not provide dense complete 3D occupancy ground truth for every sample.
- The target is a proxy built from in-painted depth, not full hidden geometry.
- The model predicts binary occupancy, not semantic labels.
- The model is deliberately lightweight and should not be described as leaderboard SOTA.
- Single-view depth observes visible surfaces only, so completion is limited by proxy-supervision quality.

9. Conclusion

This coursework project provides a full real-data pipeline for SOTA-inspired 3D scene completion using NYUv2 depth maps. It demonstrates data loading, geometric projection, voxelization, proxy target construction, volumetric neural prediction, metric evaluation, and visualization. The final system is best described as a real-data prototype rather than a benchmark-level SOTA method.

References

1. Nathan Silberman, Pushmeet Kohli, Derek Hoiem, and Rob Fergus. Indoor Segmentation and Support Inference from RGBD Images. ECCV 2012.
2. NYU Depth Dataset V2 official page: https://cs.nyu.edu/~fergus/datasets/nyu_depth_v2.html
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